Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/14/2006
PI Name:	Sacksteder, Kurt R. Ph.D.		
Project Title:	Combustion in Artificial Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Kurt.R.Sacksteder@nasa.gov	**Fax:**	FY 216-433-8660
PI Organization Type:	NASA CENTER	**Phone:**	216-433-2857
Organization Name:	NASA Glenn Research Center at Lewis Field		
PI Address 1:	MS 77-5		
PI Address 2:	21000 Brookpark Road		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Sacksteder, Kurt R	**Contact Phone:**	216-433-2857
Contact Email:	Kurt.R.Sacksteder@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ti'en, James (Case Western Reserve University) Ferkul, Paul (NASA Glenn Research Center)		
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Performance Goal No.:			
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	Partial Gravity is the environment in which planetary exploration is to occur, yet little has been done to characterize the behavior of fluid, thermal and reacting systems under reduced, but non-zero, gravitational conditions. In the case of fire safety systems, simple laboratory fuels have been shown to be substantially more flammable in partial-gravity environments than in normal Earth gravity. Such experiments are limited to the few seconds duration available using aircraft flying parabolic trajectories. While more can be done using the aircraft capability, extended duration tests will be needed to adequately explore partial-gravity behavior, especially for realistic materials and fire fighting schemes. With the exception of conducting tests on other planetary bodies, the only extended partial-gravity test capability currently envisioned is artificial gravity obtained using a low-speed centrifuge operated in an orbiting spacecraft such as the International Space Station. In contemplating using a centrifuge to obtain artificial gravity for fire related research,		

Task Description:	the Coriolis force will be present and will induce additional complexity into the behavior of flames and other fluid systems. Careful theoretical work accompanying artificial gravity experiments will relate the results to the genuine partial-gravity environment of planetary destinations, where Coriolis effects would be negligible. However, such work will also directly model behaviors within rotating spacecraft that are envisioned for long interplanetary missions to avoid the undesirable human physiology effects of microgravity. Before proposing to use a space-based centrifuge for fire research, ground-based facilities must be used to define meaningful flight test research. The purpose of this proposed ground-based research effort is to begin and stimulate the use of centrifuge facilities for understanding flame and fire behavior in an artificial gravity. The effort consists four progressive parts: 1) participation in the design and construction of a partial-gravity centrifuge for use in the NASA ground-based reduced-gravity facilities; 2) a pilot study of a stabilized stationary diffusion flame in artificial partial gravity; 3) a more extensive study of a spreading flame over a solid fuel in artificial partial gravity in which Coriolis effects will be quantified; and 4) the development of an unsteady, three-dimensional model that includes the Coriolis force. In keeping with the practical objectives of direct support for fire safety in spacecraft flown beyond Earth orbit, the flame spreading research will extend current understanding into the use of non-standard habitable atmospheres proposed for such spacecraft and the use of proposed extinguishing agents as atmospheric diluents.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated: 02/27/2004)